



Deep Learning Inference with FPGAs

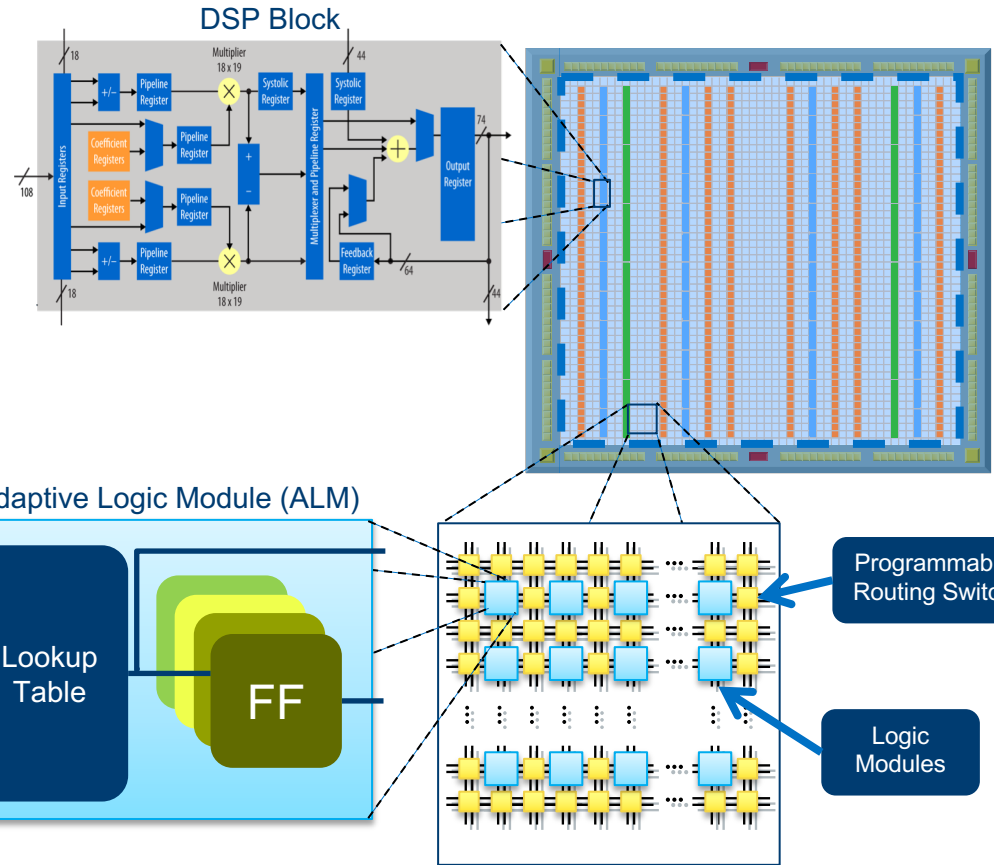
FPGA Architecture

■ Massive Parallelism

- Millions of logic elements
- Thousands of embedded memory blocks
- Thousands of Variable Precision DSP blocks
- Programmable routing
- Dozens of High-speed transceivers
- Various built-in hardened IP

■ FPGA Advantages

- **Custom hardware!**
- Efficient processing
- Low power
- Ability to reconfigure
- Fast time-to-market



Solving Machine Learning Challenges with FPGA



EASE-OF-USE

SOFTWARE ABSTRACTION,
PLATFORMS & LIBRARIES

Intel FPGA solutions enable software-defined programming of customized machine learning accelerator libraries.



REAL-TIME

DETERMINISTIC
LOW LATENCY

Intel FPGA hardware implements a deterministic low-latency data path unlike any other competing compute device.

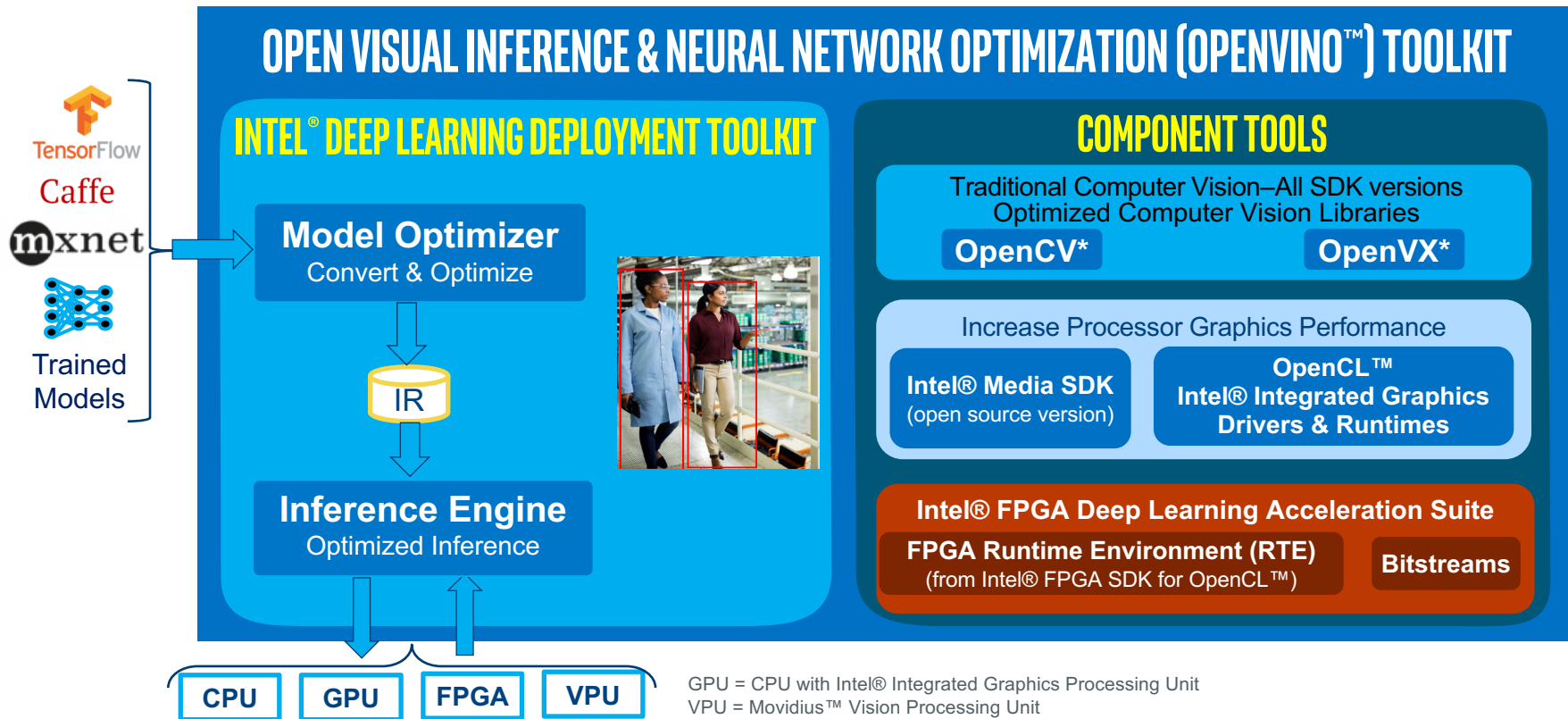


FLEXIBILITY

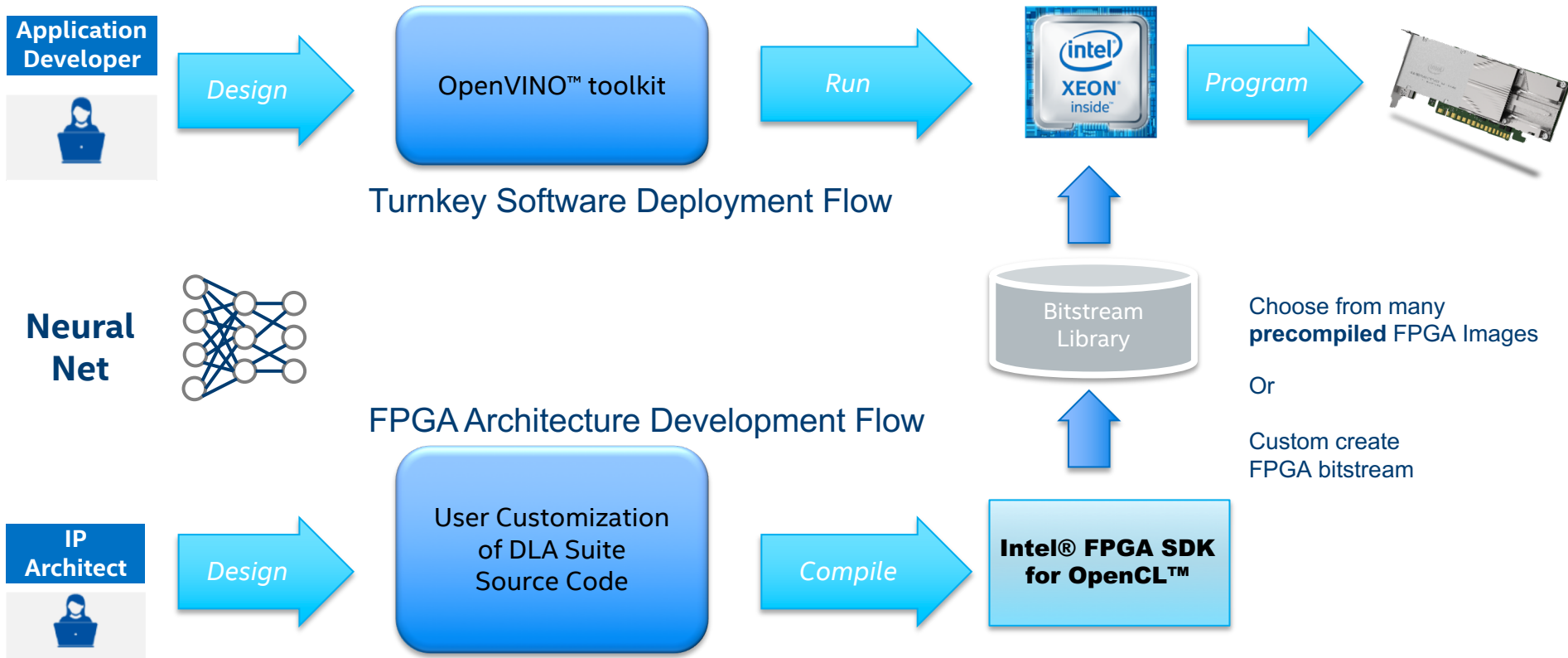
CUSTOMIZABLE HARDWARE
FOR NEXT GEN DNN ARCHITECTURES

Intel FPGAs can be customized to enable advances in machine learning algorithms.

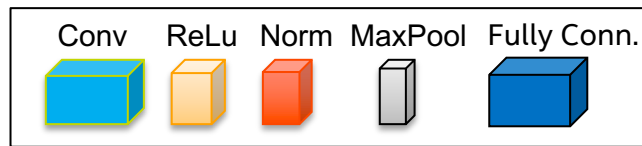
OpenVINO™ Toolkit



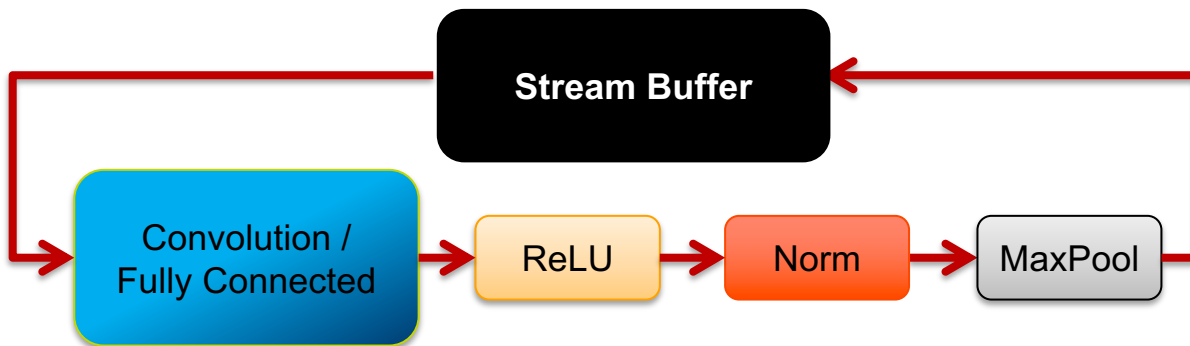
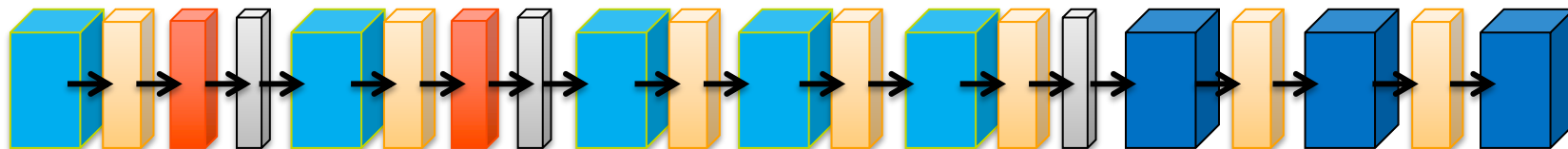
OpenVINO™ with DLA User Flows



Mapping Graphs in DLA



AlexNet Graph



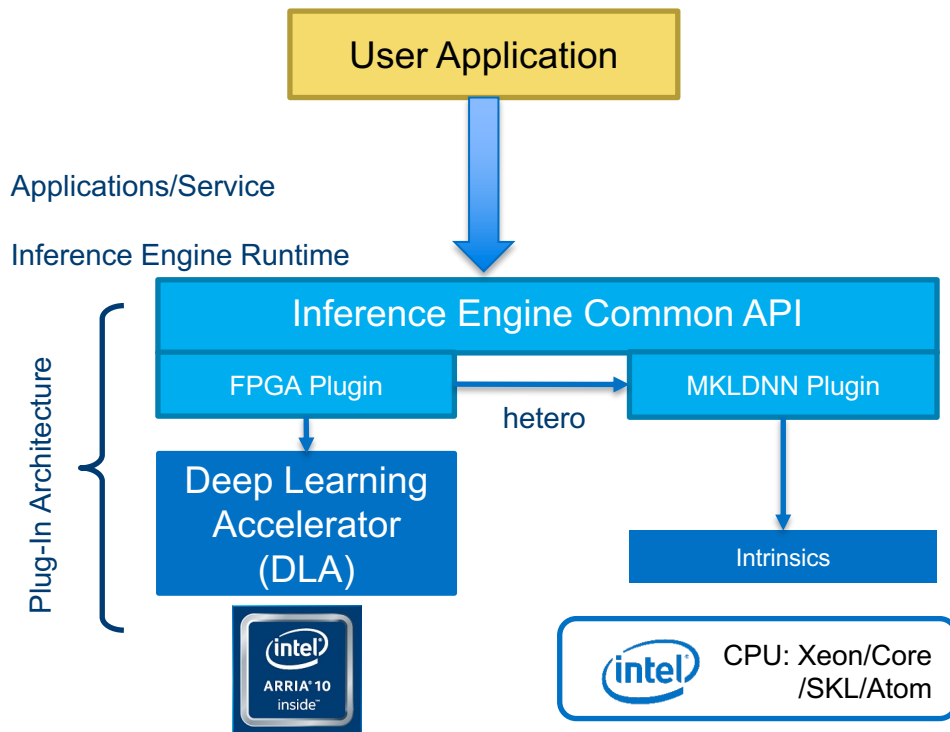
Blocks are run-time reconfigurable and bypassable

DLA Architecture Selection

- Find ideal FPGA image (arch) that meets your needs
 - Right combination of data type support, primitives support, and parallelism
- Optional: Create custom FPGA image based on need

Network	Bitstreams (Programmable Acceleration Card with Intel® Arria® 10 GX FPGA)
AlexNet	arch2, arch3, arch10, arch11, arch12, arch16, arch17, arch23, arch25, arch26
GoogleNet v1	arch2, arch3, arch11, arch12, arch16, arch17, arch23, arch25, arch26
VGG-16, VGG-19	arch2, arch3, arch16
SqueezeNet v1.0, v1.1	arch2, arch3, arch9, arch11, arch12, arch16, arch17, arch18, arch20, arch23, arch25, arch26
ResNet-18	arch2, arch3, arch9, arch16, arch20
ResNet-50	arch3, arch9, arch20
ResNet-101	arch3, arch9, arch20

Software stack



FPGA implementation is “Deep Learning Accelerator” (DLA)

Additional step: FPGA RTE (aocl) loads bitstream with desired version of DLA to FPGA before running

FPGA Supported Primitives

- The following layers are supported by the DLA plugin:

✓ Batch norm	✓ Concat	✓ Convolution
✓ Fully Connected	✓ ReLU, leaky ReLU	✓ LRN normalization
✓ Pooling	✓ ScaleShift	✓ Power
✓ Eltwise	✓ prelu	

- Heterogeneous execution
 - In case when topology contains layers not supported on FPGA, you need to use Heterogeneous plugin with dedicated fallback device.

Automatic Fallback with Hetero Plugin

```
$ classification_sample -d HETERO:FPGA,CPU ...
```

```
InferenceENgine::InferenceEnginePluginPtr enginePtr;  
enginePtr = dispatcher.getPluginByDevice("HETERO:FPGA,CPU");
```

- The “priorities” define search order
- Keeps all layers that can be executed on the device (FPGA)
- Carefully respecting the topological and other limitations
- Then follows priorities when searching (e.g. CPU)

Prepare FPGA Environment for Inference

- Intel® FPGA Runtime Environment for OpenCL™
- Prepare FPGA Board for OpenCL
- Set environment
 - Use script to ensure DLA and OpenCL libraries part of LD_LIBRARY_PATH

```
[xkqi@centos-z620 ~]$ aocl diagnose
aocl diagnose: Running diagnose from /opt/intelFPGA_pro/17.0/hld/board/a10_ref/linux64/libexec

----- aocl -----
Vendor: Intel(R) Corporation

Phys Dev Name  Status  Information
-----
acla10_ref0    Passed  Arria 10 Reference Platform (acla10_ref0)
                                   PCIe dev_id = 2494, bus:slot.func = 04:00.00, Gen3 x8
                                   FPGA temperature = 68.5547 degrees C.

DIAGNOSTIC_PASSED
-----
```

Load FPGA Image and Execute IE Application

- FPGAs needs to be preconfigured with primitives prior to application execution
- Choose FPGA bitstream from the DLA suite
 - Based on topology needs and data type requirements
 - Option to create custom FPGA bitstream based on requirements

```
-bash-4.2$ aocl program aocl0 $DLA_AOCX
aocl program: Running program from /opt/altera/aocl-pro-rte/aclrte-linux64/board/a10_ref/linux64/libexec
Programming device: a10gx : Arria 10 Reference Platform (acla10_ref0)
Reprogramming device [0] with handle 1
Program succeed.
```

- Execute User or Example Application

```
-bash-4.2$ classification_sample -d HETERO:FPGA,CPU -i car.png -m ir/squeezenet1.1/squeezenet1.1.xml
```

Summary

- FPGAs provide a flexible, deterministic low-latency, high-throughput, and energy-efficient solution for accelerating AI applications
- Intel® FPGA DLA Suite supports CNN inference on FPGAs
- Accessed through OpenVINO™ toolkit
- Available for Intel® Programmable Acceleration Card

